Max Dubowski

Edinburgh, UK · maxdubowski1112@gmail.com · maxdubowski.com
linkedin.com/in/maksymilian-dubowski · github.com/Max11122006

Education

Heriot-Watt University — Edinburgh, UK
MEng (Hons) Aerospace Engineering — *Sep 2024 – May 2028*

- Awarded the Deputy Principal's Award for achieving an 89% overall average (straight Grade A profile) across all Year 1 modules.
- Relevant Quantitative Coursework: Mathematics for Engineers, Fluid Dynamics, Thermodynamics, Finite Element Analysis (FEA), Flight Mechanics.
- International Exchange: Completed Semester 1 of Year 2 at Heriot-Watt University Malaysia (Putrajaya, Sep 2025 – Dec 2025), executing core aerospace engineering coursework within cross-campus project teams.

Work Experience

Packsync (packsync.co.uk) — Edinburgh, UK
Technical Co-Founder & Software Engineer — *Sep 2026 – Present*

- Co-founded a B2B compliance SaaS platform targeting an addressable market of 6,000 UK beverage producers, onboarding 8 pilot businesses for pEPR and DRS regulatory reporting.
- Engineered a real-time packaging fee and material stream calculation engine projected to save over 80,000 hours of manual compliance administration across target SMEs.
- Built and deployed a low-latency web application and automated client onboarding workflow, streamlining complex producer data ingestion and recurring compliance reporting.

Private Residential Portfolio — Edinburgh, UK
Assistant Property Manager (Self-employed) — *Feb 2025 – Jan 2026*

- Managed end-to-end operations, tenant check-in/out logistics, and occupancy turnover across an 8-unit residential portfolio generating £70,000 in annual rental income.
- Oversaw property maintenance budgets, scoped repair works, and negotiated with third-party trade contractors to minimize unit downtime and control operating costs.

Extracurricular Activities

BACKED – AI Prediction Market Engine | *Python, Gemini API, WebSockets, Multimodal* — Jul 2026

- Placed 2nd out of 20 competing teams at Build with Gemma: GDGoC Aberdeen, architecting a multi-threaded Python backend for a live social prediction market on study commitments.
- Implemented a probabilistic pricing model over historical user completion distributions, pairing dynamic idle-time odds drift with deterministic payout settlement logic.
- Engineered concurrent request decoupling and SHA-256 input-hash caching, preventing thread blocking during multimodal image verification and eliminating redundant inference latency.

Friendly – Real-Time Campus Coordination App | *Next.js, TypeScript, Firebase* — 2026

- Placed 5th out of 30 teams at RGU Hack 2026, building a real-time cohort scheduling platform to automate study group coordination.
- Designed an $O(N \log N)$ sweep-line interval algorithm to process multi-user availability blocks in milliseconds, surfacing optimal overlapping study slots without manual polling.

Missile Trajectory & Guidance Simulator | *Python, NumPy, Matplotlib* — 2026

- Developed a 3-DOF flight dynamics simulator modeling guided missile trajectories via 4th-order Runge-Kutta (RK4) numerical integration at a 0.01s time step.
- Integrated non-linear aerodynamic drag curves, variable mass burn, and proportional navigation guidance laws, achieving target interception within a <0.5m miss-distance tolerance.

Structural Finite Element Analysis Tool | *C++* — 2025

- Programmed a standalone matrix solver computing shear force, bending moments, and deflection profiles for statically indeterminate beam structures.
- Implemented Gaussian elimination boundary-condition solvers executing in <5ms with <0.2% deviation against analytical Euler-Bernoulli solutions and Abaqus FEA benchmarks.

Additional Information

Spoken Languages: English (Native), Polish (Native)
Technical Skills: Python, C++, SQL, MATLAB, ANSYS Fluent (CFD), Abaqus FEA
Interests: Formula 1 (attended F1 75 Live launch event in London), Golf (20 handicap; 8-year consecutive attendee at the Genesis Scottish Open), Chess (1200 Elo)